Tianjun Gan

Orcid: 0000-0002-4503-9705 Email: gtj18@tsinghua.org.cn Website: tianjungan.github.io

Education

2026.04 – now	Postdoctor Researcher, Instituto de Astrofísica de Canarias (IAC), Spain
2023.06 – 2026.04	Research Associate, Tsinghua & Westlake University, China
2018.09 – 2023.06	PhD in Astronomy, Tsinghua University, Beijing, China
2014.09 – 2018.06	BS in Physics, Zhejiang University, Zhejiang, China

Research Interests

Formation and evolution of planets around M-type stars;
Planet atmospheres through high-resolution transmission spectroscopy;
Characterization and population statistics of BDs and low-mass M dwarfs;
Connection between stellar abundance pattern and planet formation, esp. solar twins;
Transiting planet detection, follow-up and characterization.

Research Programs

GPASS (Giant Planets Around Small Stars) program, Lead	2019-Present;
TESS low mass stellar companion program, Lead	2019-Present;
SPEAR Project (PI: Enric Pallé), Member	2026-Present
LCO Key Followup Program for TESS (PI: Avi Shporer), Member	2019-Present;
Magellan TESS Survey (PI: Johanna Teske), Member	2020-Present;
TESS Follow-up Observing Program (TFOP), Member	2019-Present.

Observing Proposals & Experience

As PI or Science PI:

2026B	14.0 hours on the CFHT (SPIRou);
2026B	8.0 hours on the Gemini-North (MAROON-X);
2026A	4.7 hours on the TNG (HARPS-N); spectroscopy–*First transmission spectroscopy study of an ultra-hot Jupiter with the fourth-highest equilibrium temperature*
2026A	7.0 hours on the CFHT (SPIRou); spectroscopy–*RV Confirmation for the third warm Jupiter around an M star*
2025B DDT	8.8 hours on the CFHT (SPIRou); spectroscopy–*RV monitoring for the first M dwarf giant planet system with TTV signals*
2025A	4.8 hours on the CFHT (SPIRou); spectroscopy–*First obliquity measurement for an M dwarf hosting a very-low-mass star*
2024B	3 hours on the CAHA (CARMENES); spectroscopy– *Dawn of the Gaia era: validation for three astrometric planet candidates*
2024B	6 hours on the CFHT (SPIRou); spectroscopy– *Mass determination for a rare warm Jupiter candidate orbiting an M star*

2024A DDT	6.0 hours on the CFHT (SPIRou); spectroscopy– *RV monitoring for the first hot+warm Jupiter system around an M dwarf that favors gravitational instability models*
2024A DDT	1.5 hours on the CFHT (SPIRou); spectroscopy– *RV confirmation for a hot Jupiter candidate around an M4.5V dwarf that challenges core accretion models*
2024A	7.5 hours on the CFHT (SPIRou); spectroscopy– *Mass determination of two transiting warm Jupiter candidates around M dwarfs*
2023B Fast-Turnaround	5.3 hours on the Gemini-North (MAROON-X); spectroscopy– *First attempt to measure the obliquity of an M dwarf hosting a hot Jupiter*
2023B	9 hours on the AAT (Veloce); spectroscopy– *Recon spectroscopic observations for TESS planet candidates around metal-poor stars*
2023B	15 hours on the IRTF (SpeX); spectroscopy– *Homogeneous stellar characterization for M dwarfs with confirmed giant planets*
2023A	1.5 night on the CFHT (SPIRou); spectroscopy– *Mass measurement of a planet candidate that challenges planet formation models*
2022B	1.5 night on the CFHT (SPIRou); spectroscopy– *Mass measurement of a hot Jupiter around an M dwarf delivered by TESS*
2022B	1 night on the Xinglong 2.16m telescope; spectroscopy – *Rossiter-McLaughlin observation for TOI-1830: An eccentric low mass stellar companion around a young star*
2022A	60 hours on SMARTS 1.5-m Telescope (CHIRON); spectroscopy – *Investigating the solar depletion pattern with TESS solar analogs*
2021A	2 night on the CFHT (SPIRou); spectroscopy – *Mass determination for a planet around an M dwarf close to the radius valley*
2021A	3 nights on the LCOGT network (1m0 Sinistro); photometry – *Follow-up observations for TESS planet candidates*
2020A	1 night on the CFHT (SPIRou); spectroscopy – *Confirmation of the sixth transiting giant planet around an M dwarf*
2020A	7.5 nights on the LCOGT network (1m0 Sinistro); photometry – *Photometric followup observations for TESS hot Jupiters around M dwarfs*
As Co-I:	
2025B	17.5 hours on the CAHA (CARMENES); spectroscopy– *Measuring the Masses of Three Transiting Giant Planet Candidates Around M Dwarfs with CARMENES* (PI: Felipe Murgas)
2024B	14 hours on Magellan/PFS; spectroscopy – *Obliquity determination for Au Mic b&c* (PI: Zitao Lin)
2024B	3 nights on LBT; spectroscopy – *Abundance measurements for planets around solar twins/analogs (PASTA)* (PI: Qinghui Sun)
2022A	4 nights on LCOGT network (1m0 NRES); spectroscopy – *Radial Velocity Follow-ups of TESS Discovered Small Planets to Search for Additional Gas Giants* (PI: Xinyan Hua)
2020A-2021B	10/800/400 hours on LCOGT 2m0/1m0/0m4 telescopes; photometry – *Coordinated photometric follow-up of TESS candidates* (PI: Karen A. Collins)

2020-now	100/1000/800 hours each semester on LCOGT 2m0/1m0/0m4 telescopes; photometry+spectroscopy – *Standing on the shoulders of the network: Follow-up of TESS planet candidates with LCO* (key proposal, PI: Avi Shporer)
2019B	10/180/360 hours on LCOGT 2m0/1m0/0m4 telescopes; photometry – *Coordinated photometric follow-up of TESS candidates* (PI: Markus Rabus)

Publication List

16 as the first/second author
*** = student co-supervised**

Leading Author: ADS Library

1. **Gan, T.** & L'Heureux, A. et al., *Stellar Multiplicity of M Dwarfs with Short-period Giant Planets, and the Characterization of TOI-5628Ab*, 2026, ApJL, 1006, 63
2. **Gan, T.** & L'Heureux, A. et al., *An Aligned Very-Low-Mass Star Orbiting an M dwarf and Obliquity Patterns Across Giant Planets, Brown Dwarfs, and Binary Stars*, 2026, ApJL, 1002, 8
3. **Gan, T.** & Orell-Miquel, J. et al., *Revisiting the atmosphere of HAT-P-70b with CARMENES high-resolution transmission spectroscopy*, 2026, AJ, 171, 89
4. **Gan, T.** & Cadieux, C., et al., *A New Brown Dwarf Orbiting an M star and An Investigation on the Eccentricity Distribution of Transiting Long-Period Brown Dwarfs*, 2025, ApJL, 988, 78
5. **Gan, T.** & Theissen, C., et al., *Metallicity Dependence of Giant Planets around M Dwarfs*, 2025, ApJS, 276, 47
6. **Gan, T.** & Wang, X. S., et al., *The Aligned Orbit of a Hot Jupiter around the M Dwarf TOI-4201*, 2024, ApJL, 969, 24
7. **Gan, T.** & Guo, K., et al., *Relative Occurrence Rate Between Hot and Cold Jupiters as an Indicator to Probe Planet Migration*, 2024, ApJ, 967, 74
8. **Gan, T.**, *Gaia Astrometry and MIKE+PFS Doppler Data Joint Analysis Reveals that HD 175167b is a Massive Cold Jupiter*, 2023, RNAAS, 7, 226
9. **Gan, T.** & Cadieux, C., et al., *A massive hot Jupiter orbiting a metal-rich early-M star discovered in the TESS full frame images*, 2023, AJ, 166, 165
10. Lin, Z. **(*)**, **Gan T.**, et al., *One high mass brown dwarf and two objects near the hydrogen burning mass limit*, 2023, MNRAS, 523, 6162
11. **Gan, T.** & Wang, X. S., et al., *Occurrence rate of hot Jupiters around early-type M dwarfs based on TESS Primary Mission*, 2023, AJ, 165, 17
12. **Gan, T.** & Soubkiou, A., et al., *TESS discovery of a sub-Neptune orbiting a mid-M dwarf TOI-2136*, 2022, MNRAS, 514, 4120
13. **Gan, T.** & Lin, Z. **(*)**, et al., *TOI-530b: A giant planet transiting an M dwarf detected by TESS*, 2022, MNRAS, 511, 83
14. **Gan, T.** & Bedell, M., et al., *HD 183579b: a warm sub-Neptune transiting a solar twin detected by TESS*, 2021, MNRAS, 507, 2220

15. **Gan, T.** & Wang, X. S., et al., *Revisiting the HD 21749 planetary system with stellar activity modelling*, 2021, MNRAS, 501, 6042
16. **Gan, T.** & Shporer, A., et al., *LHS 1815b: The First Thick-disk Planet Detected by TESS*, 2020, AJ, 159, 160

Selected Contributed Work: (see the full list of 96 publications here: ADS Library)

1. Liu, Q., Zhu, W., **Gan T.**, et al., *TESS planets in known radial velocity cold Jupiter systems: Hot super Earth occurrence is enhanced by cold Jupiters*, 2026, AJ, 172, 72
2. Zhai, R., Lee, M. H., **Gan T.**, Mao, S., et al., *Dynamical Instability of Multiplanet Systems and Free-floating Planets*, 2025, ApJ, 990, 74
3. Sun, Q., Wang, X. S., **Gan T.**, et al., *Planets Around Solar Twins/Analogs (PASTA) I.: High precision stellar chemical abundance for 17 planet-hosting stars and the condensation temperature trend*, 2025, ApJ, 980, 179
4. Yang, H., Zang, W., **Gan T.**, et al., *How Rare Are TESS Free-floating Planets*, 2024, ApJL, 972, 12
5. Hua, X., Wang, X. S., Teske, Johanna K., **Gan T.**, et al., *A Transiting Super-Earth in the Radius Valley and an Outer Planet Candidate Around HD 307842*, 2023, AJ, 166, 32
6. Sun, Q., Wang, X. S., **Gan T.**, et al., *A Search for Exoplanets in Open Clusters and Young Associations based on TESS Objects of Interest*, 2022, RAA, 22, 7
7. Teske, J., Wang, X. S., Wolfgang, A., **Gan, T.**, et al., *The Magellan-TESS Survey. I. Survey Description and Midsurvey Results*, 2021, ApJS, 256, 33
8. Bryant E., et al. (incl. **Gan, T.**), *A transiting giant planet in orbit around a 0.2-solar-mass host star*, 2025, Nature Astronomy, 9, 1031
9. Armstrong D., et al. (incl. **Gan, T.**), *A remnant planetary core in the hot-Neptune desert*, 2020, Nature, 583, 39
10. Günther M., et al. (incl. **Gan, T.**), *A super-Earth and two sub-Neptunes transiting the nearby and quiet M dwarf TOI-270*, 2019, Nature Astronomy, 3, 1099
11. Vanderspek R., et al. (incl. **Gan, T.**), *TESS Discovery of an Ultra-short-period Planet around the Nearby M Dwarf LHS 3844*, 2019, ApJ, 871, 24

Selected Seminar and Conference Talks, Posters

2026.08	Seminar at Université de Montréal (Invited Talk)
2026.06	CSST/CPIC Cool Planets and Disks (Invited Talk)
2025.12	Exoplanet and Planet Formation Conference (Poster)
2025.10	CARMENES consortium meeting (Contributed Talk)
2025.08	Earth 2.0 (ET) Space Mission Annual Science Meeting (Contributed Talk)
2025.07	CSST (Chinese Space Station Survey Telescope) Science Conference (Contributed Talk)
2025.06	Seminar at Nanjing University (Seminar)
2024.11	High Precision Astrometry Post-Gaia (Contributed Talk)
2024.07	TESS Science Conference III (Poster)
2024.06	Exoplanet V (Poster)
2023.12	Open Problems in the Astrophysics of Gas Giants (Contributed Talk)
2023.10	Exoplanet and Habitable Worlds seminar at Penn State (Seminar)

2023.10 TESS Science Talk at MIT (Seminar)
2023.10 SPLAT talk at University of Hawaii (Seminar)
2023.08 Asia Oceania Geosciences Society 2023 (Contributed Talk)
2023.04 2023 International Conference of Deep Space Sciences (Contributed Talk)
2023.03 The 5th Youth Planet Conference (Contributed Talk)
2022.12 Earth 2.0 (ET) Mission Science Seminar (Invited Talk)
2022.11 Caltech: *The mysteries of giant planets around M dwarfs* (Group Meeting Talk)
2022.10 TESS Science Team Meeting #29 (Contributed Talk)
2022.07 Cool Stars 21 conference (Poster)
2022.01 CFHT/SPIRou Science Seminars (Invited Talk)
2021.12 The China's Telescope Access Program meeting (Virtual)
2021.12 Chinese Astronomical Society Meeting (Contributed Talk)
2021.08 TESS Science Conference II (Contributed Talk)
2021.06 Chinese Planetary Science Society Annual Conference (Contributed Talk)
2020.12 Earth 2.0 (ET) Mission workshop (Invited Talk)
2020.12 Earth 2.0 (ET) Transit Space Mission Science Meeting (Invited Talk)

Teaching and Mentoring Experience

2022 Spring TA for Observational Astronomy (Instructor: Prof. Xuesong Wang)
2020 & 2021 Fall TA for The Beauty of the Universe (Instructor: Prof. Shude Mao)

2025 Lei Ren, Undergraduate at Westlake University, China.
2022 Ximing Xu, Undergraduate at Western University, Canada; TFOP member.
2020 – 2022 Zitao Lin, Undergraduate at Tsinghua University, China; Now PhD candidate at Tsinghua University, China; TFOP member.
2020 – 2021 Gavin Wang, High school student from Tsinghua International School and Stanford Online High School, China; Now undergraduate student at Johns Hopkins University, USA; TFOP member.

Professional Services

Referee for A&A, AJ, ApJ, PASA.
Reviewer for CFHT proposals
Reviewer for VLT proposals
Reviewer for Gemini FT proposals

Awards

2024 Trottier Postdoctoral Fellowship at Université de Montréal
2024 Sullivan Postdoctoral Fellowship at Indiana University (declined)
2023 VIDA Postdoctoral Fellowship at Vanderbilt University (declined)
2023 Jiang Nanxiang Scholarship, Tsinghua University
2022 Second-class Scholarship of China Astronomical Society
2021 National Scholarship, Tsinghua University (*Highest Student Award*)
2020 First-class Academic Scholarship, Tsinghua University
2019 First-class AMD Scholarship, Tsinghua University

2017 Second-class Academic Scholarship, Zhejiang University
2016 National Scholarship, Zhejiang University (*Highest Student Award*)
2015 National Encouragement Scholarship, Zhejiang University
2015 First-class Academic Scholarship, Zhejiang University

VISITING EXPERIENCE

- 2023.12 – 2024.06: Visiting Postdoc, host: Enric Pallé, Instituto de Astrofísica de Canarias (IAC), E-38205 La Laguna, Tenerife, Spain

- 2019.10 – 2020.01: Visiting Student, host: Stephen Shectman, Observatories of the Carnegie Institution for Science, 813 Santa Barbara Street, Pasadena, CA 91101, USA